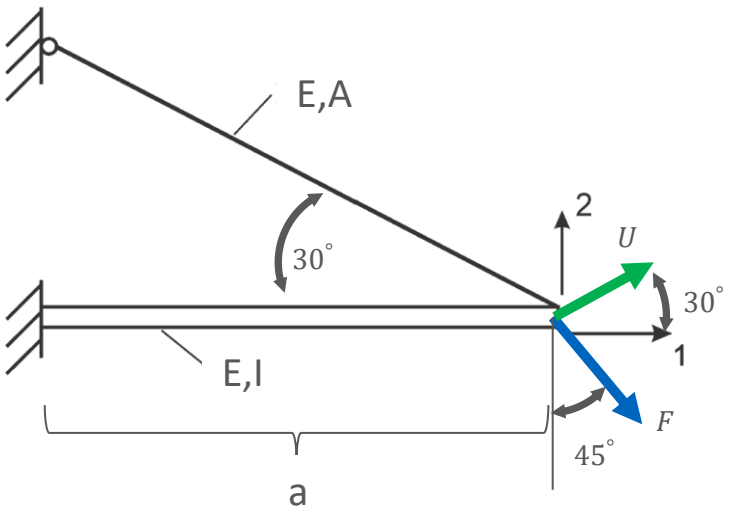
Task 1

The structure shown on the right consists of a beam and a truss and is to be analyzed using the finite element method. The two load cases **F** and U are considered separately.



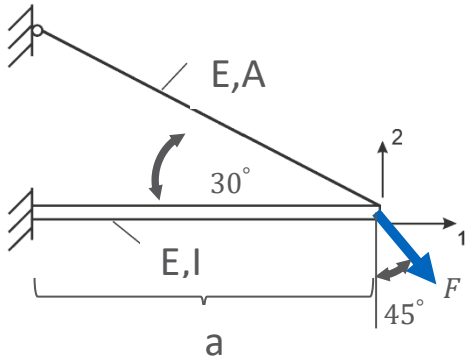
Load Case 1: F

a) Calculate the deflection of the degree of freedom 1 and 2 due to the applied load F ($\boldsymbol{EI} = 70Nm^2 \quad \boldsymbol{EA} = 700N$).
b) Calculate the stress in the truss structure.
c) Calculate the reaction forces in the support points.
d) Calculate the maximum stress in the beam Element. Assume a rectangular cross section with the hight 0.002m

Load Case 2: U

e) The displacement at the tip shall be $U = 0.5$. Calculate the necessary applied load vector F'

Task 1



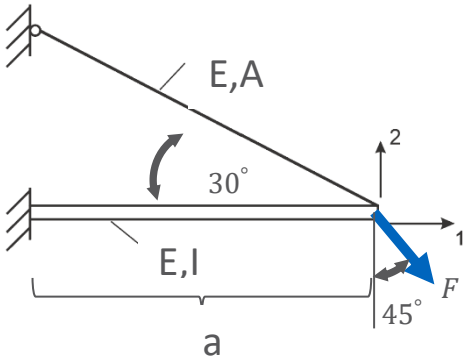
a) Calculate the deflection of the degree of freedom 1 and 2 due to the applied load F.

$$\boldsymbol{EI = 70Nm^2 \quad EA = 700N \quad F = 1N}$$

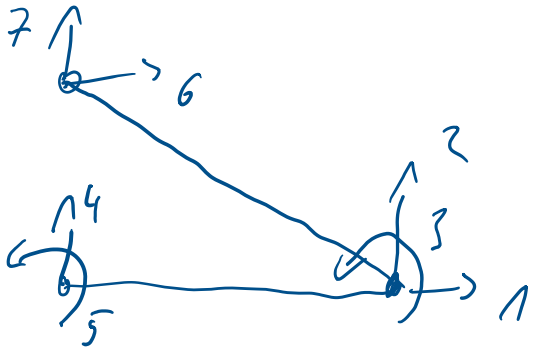
$$K = \begin{bmatrix} \frac{3}{4}\frac{EA}{l} & -\frac{\sqrt{3}}{4}\frac{EA}{l} & 0.0 \\ -\frac{\sqrt{3}}{4}\frac{EA}{l} & \frac{1}{4}\frac{EA}{l} + 12\frac{EI}{a^3} & -6a\frac{EI}{a^3} \\ 0.0 & -6a\frac{EI}{a^3} & 4a^2\frac{EI}{a^3} \end{bmatrix} = \begin{bmatrix} 454.732291 & -262.53981062 & 0.0 \\ -262.53981062 & 991.57743033 & -420.0 \\ 0.0 & -420.0 & 280. \end{bmatrix}$$

Task 1

a) Calculate the deflection of the degree of freedom 1 and 2 due to the applied load F.



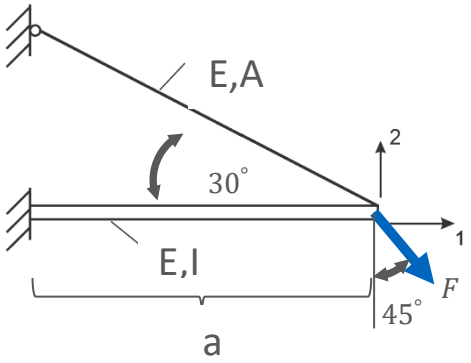
$$F = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \\ 0 \end{bmatrix}$$



$$U = \begin{bmatrix} U_1 \\ U_2 \\ U_3 \end{bmatrix} = \begin{bmatrix} 0.00073335 \\ -0.00142314 \\ -0.0021347 \end{bmatrix}$$

Task 1

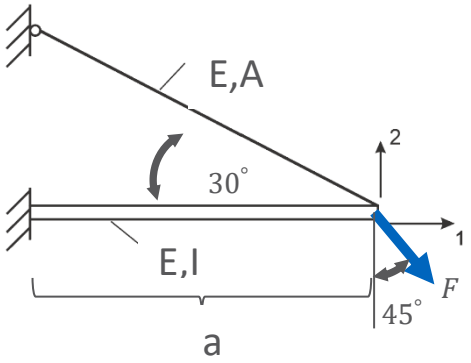
b) Calculate the stress in the truss structure.



$$\sigma = \frac{E}{l}\left(\frac{\sqrt{3}}{2}U_1 - \frac{U_2}{2}\right)$$

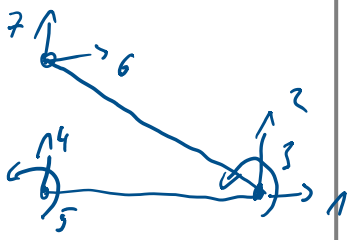
Task 1

c) Calculate the reaction forces in the support points.

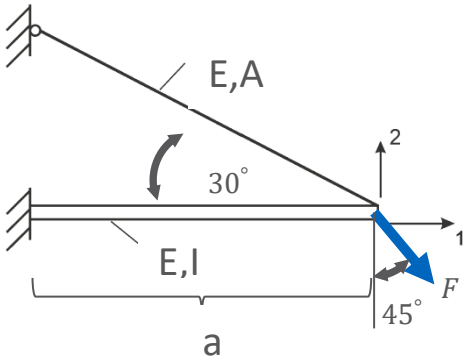


a

$$K_G = \begin{bmatrix} 4 & 5 & 6 & 7 & 1 & 2 & 3 \\ X & X & X & X & 0 & -12\frac{EI}{a^3} & 6a\frac{EI}{a^3} \\ & X & X & X & 0 & -6a\frac{EI}{a^3} & 2a^2\frac{EI}{a^3} \\ & & X & X & -\frac{3}{4}\frac{EA}{l} & \frac{\sqrt{3}}{4}\frac{EA}{l} & 0 \\ & & & X & \frac{\sqrt{3}}{4}\frac{EA}{l} & -\frac{1}{4}\frac{EA}{l} & 0 \\ & & & & X & X & X \\ & & & & X & X & X \\ & & & & X & X & X \end{bmatrix} \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ U_1 \\ U_2 \\ U_3 \end{Bmatrix} \begin{matrix} 4 \\ 5 \\ 6 \\ 7 \\ 1 \\ 2 \\ 3 \end{matrix} = \begin{Bmatrix} 0.299061 \\ 0.299061 \\ -0.707 \\ 0.407939 \\ X \\ X \\ X \end{Bmatrix}$$



Task 1



d) Calculate the maximum stress in the beam Element.

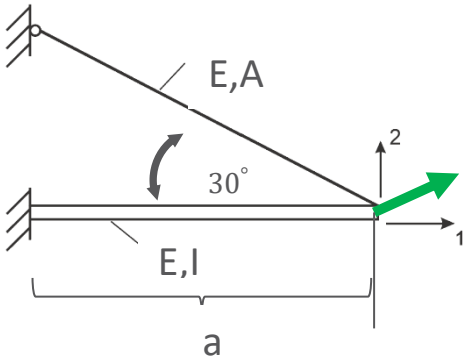
$$\boldsymbol{I = 0.001m^4}$$



$$\sigma_{max} = \frac{M_{max}}{I} z_{max} = \frac{0.299061\ Nm}{0.001m^4} 0.001m = 2.99061 \cdot 10^{-4}\ MPa$$

Task 1

e) The displacement at the tip shall be $U = 0.5$. Calculate the necessary applied load vector F'



$$\begin{bmatrix} \frac{3}{4}\frac{EA}{l} & -\frac{\sqrt{3}}{4}\frac{EA}{l} & 0.0 \\ -\frac{\sqrt{3}}{4}\frac{EA}{l} & \frac{1}{4}\frac{EA}{l} + 12\frac{EI}{a^3} & -6a\frac{EI}{a^3} \\ 0.0 & -6a\frac{EI}{a^3} & 4a^2\frac{EI}{a^3} \end{bmatrix} \begin{Bmatrix} \frac{\sqrt{3}}{4} \\ \frac{1}{4} \\ U_3 \end{Bmatrix} = \begin{Bmatrix} F_1 \\ F_2 \\ 0 \end{Bmatrix}$$

$$F_1 = \frac{3}{4}\frac{EA}{l}\frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4}\frac{EA}{l}\frac{1}{4}$$

$$F_2 = -\frac{\sqrt{3}}{4}\frac{EA}{l}\frac{\sqrt{3}}{4} + \left(\frac{1}{4}\frac{EA}{l} + 12\frac{EI}{a^3}\right)\frac{1}{4} - 6a\frac{EI}{a^3}U_3$$

$$0 = -6a\frac{EI}{a^3}\frac{1}{4} + 4a^2\frac{EI}{a^3}U_3$$